

Which Damage Do the Images Support? Evidence-Bounded 3D Damage Estimation on Turbine Blades

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Abstract

Visual inspection of aero-engine blades must decide whether a change seen in an image is physical damage, where it is on the part, and how deep it goes. Existing methods flag anomalies in 2D, output boxes or voxels, or require a measured 3D scan, and none states whether the available views are sufficient to support a claimed shape change. We study evidence-bounded 3D damage estimation: given a healthy CAD model and a few calibrated images, output a displacement field on the blade surface when the images support one, and abstain when camera error, lighting or a surface-colour change explain the images equally well. Our estimator compares a nuisance-only explanation (per-view pose, light gains, an albedo patch) with the same explanation plus a continuous dent, using a differentiable renderer with a neural shader. Dent hypotheses are proposed by a Rao score test computed for every surface location in one backward pass, and the decision threshold on the resulting fit gain is calibrated on healthy and colour-only blades with a split-conformal guarantee. We release BladeDelta, a benchmark of [TODO: N] rendered inspection cases on [TODO: N] public Rotor37 blade geometries with exact damaged meshes, colour-only negatives, three dent families and controlled capture perturbations, with a sealed test split. [TODO: one sentence with headline test numbers]

1. Introduction

Turbine and compressor blades are inspected with cameras, and the inspector's question is physical: is there a dent, where on the blade is it, and how deep is it? Damage of interest is small. On a blade imaged at 512×512 pixels, a dent of 0.07 mm changes a few hundred pixels by two to seven grey levels. A one-degree camera error, a ten-percent lighting change or a stain changes thousands.

Current vision methods answer an easier question. Industrial anomaly detection localises unusual pixels [1, 18], including under pose and illumination change [3, 11]. Reference-based methods compare a photograph with a CAD model and output boxes [2], or lift per-view scores into an anomaly volume [7]. CAD-conditioned reconstruction fits a deformation field to sparse posed views [17, 21], and leaves measurement-quality damaged surfaces to future work [17]. None of these outputs states whether the images actually support a physical change, as opposed to a nuisance explanation.

Our own earlier experiments made this concrete. A render-and-compare fit on public Rotor37 blade geometry produced false dents on 14 of 14 healthy or colour-only blades under a one-degree camera error, and on unfamiliar dents the best image match selected a worse 3D shape in 8 of 12 depth sweeps. Better image agreement is not better geometry, and a plausible reconstruction can choose the wrong physical explanation.

We therefore formulate evidence-bounded 3D damage estimation. The input is a healthy CAD mesh and K calibrated images; the output is a per-vertex displacement field together with a decision, supported or not supported by these views. Our estimator treats this as a nested hypothesis comparison. H_0 explains the images with nuisance only: per-view pose corrections, light gains and a surface albedo patch. H_1 adds a continuous dent. The fit gain $L(H_0) - L(H_1)$ is thresholded at a value calibrated on negatives, giving a controlled false-change rate.

Making this work required separating the roles of the evidence. Pose is fitted from silhouettes, which do not depend on lighting; the dent is compared against nuisance only on interior surface pixels that are not grazing, not near a silhouette and not on the blade root. Dent hypotheses come from a Rao score test evaluated for every surface location and radius with one backward pass through the renderer, and the dent and colour-patch explanations are searched symmetri-

076 cally, so a colour patch fitted first cannot hide a dent.
077 We also report observability: on our development split,
078 70% of dents placed on camera-facing surfaces are less
079 than half visible in the two input views, and the correct
080 output for those is to abstain.

081 Contributions:

- 082 • The task of evidence-bounded 3D damage estimation
083 from sparse views and healthy CAD, with metrics
084 that separate localisation, shape accuracy, false
085 physical change and abstention on unobservable
086 damage.
- 087 • An estimator combining a differentiable renderer
088 with a neural shader, silhouette-only pose, interior-
089 domain nested hypothesis testing, score-test proposals
090 and split-conformal calibration.
- 091 • BladeDelta, a benchmark built on public Rotor37
092 geometry [4] with exact damaged meshes, colour-
093 only negatives, three dent families that differ from
094 the estimator’s own profile, and controlled camera
095 and lighting perturbations, with a protocol and
096 sealed test split fixed before evaluation.

097 2. Related Work

098 Industrial and 3D anomaly detection. Memory-bank
099 methods such as PatchCore [18] localise anomalous pixels;
100 MVTec 3D-AD [1] and Anomaly-ShapeNet [13]
101 add measured or synthetic 3D. Pose- and illumination-
102 agnostic settings use Gaussian-splat references [11]
103 [TODO: cite PIAD after verifying the primary source],
104 and M²AD [3] benchmarks view–illumination inter-
105 play. Multi-view settings include RAD [5] and
106 SiM3D [7], which lifts per-view scores into an anomaly
107 volume; recent work warns that adding views can leak
108 information in multi-view evaluation [6]. Synthetic
109 anomalies have been argued sufficient for 3D detection
110 [20], and incomplete-modality methods train with
111 3D and test on RGB [19]. These methods output
112 anomaly scores; they do not estimate a physical surface
113 change or state whether the views support one.

114 Reference CAD and blade inspection. Looking 3D [2]
115 aligns a query photograph with a reference shape and
116 localises abnormal regions with boxes. BladeSynth [8]
117 renders procedural nicks, dents, scratches and corrosion
118 on aero-engine blades for 2D detection. Model-free universal
119 detection reconstructs a normal shape by GAN
120 inversion [15]. We instead output a 3D displacement
121 field on a known design.

122 CAD-conditioned reconstruction. CAD-NeRF [21] re-
123 trieves a CAD model and jointly optimises deformation,
124 pose and appearance from few uncalibrated views.

CADsplat [17] fits a shared deformation field to sparse
posed views with a CAD prior and explicitly leaves
measurement-quality deformed surfaces to future work.
Our free-form deformation baseline follows this family.
The difference is the output: a decision on whether
the deformation is supported, with a calibrated false-
change rate.

Inverse rendering and uncertainty. Inverse rendering
separates shape and material from images [16, 22];
we use differentiable rasterisation [12] with a learned
shader for the same purpose on a known geometry.
Radiance-field uncertainty [9, 10] and observation completeness [14] quantify where a reconstruction is unreliable. They do not test a physical-change hypothesis against nuisance explanations, which is the question an inspector needs answered.

511 3. Method

Setup. We are given a healthy mesh $\mathcal{M} = (V, F)$ with
vertices $V \in \mathbb{R}^{N \times 3}$, K observed images $I_1, \dots, I_K$ with
nominal intrinsics and extrinsics, and nominal lighting.
We seek a displacement field $D \in \mathbb{R}^{N \times 3}$ on the
airfoil and a decision $s \in \{\text{supported}, \text{not supported}\}$.
Nuisance variables η are a per-view pose correction (a
rotation about the part and a camera-frame translation),
three light gains, and an optional albedo patch.

Differentiable renderer. We rasterise $\mathcal{M} + D$ with nvd-
iffrast [12] using smooth vertex normals. A neural
shader maps the per-pixel world normal n , position x
and view direction ω to one radiance component per
light, $R_k(n, x, \omega)$, and an environment term R_0 . Because
light transport is linear in each light’s intensity, the pixel
colour is $T(a \odot (R_0 + \sum_k g_k R_k))$ for albedo multiplier
 a , light gains g and a learned monotone tone curve
 T , followed by anti-aliasing and the sensor’s pixel filter.
An unstructured shader that took g as an input learned
to ignore it, and lighting changes then produced spurious
dents. The shader is trained once, on renders of blade
geometries disjoint from all evaluation geometries, with a
paired loss on image differences between lighting settings.
The healthy CAD rendered under the nominal capture
provides a per-pixel correction $B = I^{\text{ref}} - \hat{I}(\mathcal{M}; \eta_0)$ that
absorbs static shader error. On a held-out blade its mean
error is 0.92 grey levels; image changes caused by a lighting
change correlate with the reference renderer at 0.90–0.98,
and changes caused by a dent at 0.96–0.98 in the 120° view.

Nuisance fit (H_0). Each view’s pose is initialised by
Euclidean ECC alignment of the observed and refer-

ence images, then refined by matching anti-aliased silhouettes to the observed foreground, coarse to fine. Silhouettes do not depend on lighting, which prevents the pose from compensating for a brightness change. With pose fixed, light gains are fitted on the full image.

Evidence domain. All further comparisons use a fixed interior domain Ω : airfoil pixels at least 6 px from any silhouette or self-occlusion edge, excluding the blade root and pixels where $|n \cdot \omega| < 0.35$. Both hypotheses are evaluated on the same Ω , so their losses are comparable.

Dent model (H_1). A dent is parameterised by a sub-vertex centre c (an anchor vertex plus a tangent offset), depth $\delta \geq 0$, radius r , aspect ratio, orientation, profile exponent p and a signed rim coefficient κ :

$$D_i = -\delta \left[(1 - s_i^2)_+^p - \frac{\kappa}{2} \rho(s_i) \right] n_c, \quad (1)$$

where s_i is the anisotropic normalised distance of vertex i from c in the tangent plane, ρ a ring profile and n_c the centre normal, restricted to same-side airfoil vertices. None of the benchmark’s truth families matches this profile exactly.

Score-test proposals. Let $h \in \mathbb{R}^N$ be an inward normal displacement. One backward pass gives $g = -\partial L_\Omega / \partial h$ at $h = 0$. For a kernel $w_{c,r}$ centred at c with radius r , the normalised score $\langle w_{c,r}, g \rangle / \|w_{c,r}\|$ is the Rao score statistic for a small dent at (c, r) against H_0 . We evaluate it for all candidate centres and two radii with a sparse matrix product, and start H_1 fits from the top- k after non-maximum suppression. Proposals are computed from the nuisance-only residual, and each start is fitted both with and without the H_0 albedo patch, so a colour explanation fitted first cannot hide a dent.

Decision. The fit gain is $\Delta = L_\Omega(H_0) - L_\Omega(H_1) \geq 0$, as H_1 nests H_0 . We report D if $\Delta > \tau$ and $D = 0$ otherwise, with τ the split-conformal $(1-\alpha)$ quantile of Δ on calibration negatives (healthy and colour-only blades), which bounds the false-change rate on exchangeable negatives by α . We use a Charbonnier loss ($\epsilon = 2/255$) summed over Ω ; a Student- t noise model and a wider Charbonnier floor did not improve separation on development data.

4. The BladeDelta Benchmark

Geometry. Healthy blades are public Rotor37 compressor geometries [4] (CC BY-SA 4.0), normalised to

Split	Blades	Healthy	Stain	Dent	Dent + stain
Shader training	13	only for the renderer; never evaluated			
Development	12	12	24	36	24
Calibration	12	24	36	36	24
Test (sealed)	30	30	60	90	60

Table 1. BladeDelta case counts. Each case has 7 rendered views; methods use 2 by default.

unit span and closed with a labelled artificial hub collar. Each mesh has 29,989 vertices. Shapes 0–12 were used during earlier development and only to train the shader. Shapes 13–24 form the development split, 25–36 the calibration split and 37–66 the sealed test split; 67–109 are held in reserve.

Damage. Each blade receives one of: no change, a colour-only patch (a soft darkened or rust-tinted albedo change), a dent, or a dent plus a colour patch (half overlapping). Dents follow one of three families — a cubic profile with a raised rim, a Gaussian, and an anisotropic elliptic profile (aspect 0.4–1) — with depth log-uniform in $[0.0005, 0.006]$ blade spans (about 0.04–0.45 mm on a 74 mm Rotor37 blade) and radius in $[0.04, 0.12]$. Centres face at least one of seven cameras and lie at least $\max(0.1, 1.2r)$ from the airfoil boundary.

Capture. Images are rendered with Blender Cycles (32 samples, denoised, AgX tone mapping, 512×512) from seven cameras on a ring around the blade, under one of four conditions: nominal; camera jitter (yaw, pitch and roll each uniform in $\pm 1.5^\circ$ per view); light jitter (each of three area lights $\times [0.85, 1.15]$); or both. Methods receive only the nominal cameras and lighting. The default input is two views (60° and 120°); four- and seven-view settings are ablations.

Protocol. The split, case lists, metrics and hypotheses were hash-frozen before any test image was rendered; protocol changes are logged with dates. For each damaged case we record observability, the depth-weighted fraction of the dent visible in the input views.

CAD6 transfer split. To test transfer beyond Rotor37 we add six industrial blade designs supplied as STEP solids (airfoil spans 24–70 mm), with platforms, fillets and dovetail roots. The airfoil is the largest B-rep face (two faces on one blade); everything else is excluded from evidence like the Rotor37 collar. Each blade is normalised to unit airfoil span in the Rotor37 frame

and receives 76 cases (456 in total): 30% dents centred where both input cameras see the surface, 15% dents hidden from both, 20% healthy, 15% stain only, 10% dent plus stain and 10% healthy with camera error alone ($\pm 0.5^\circ$, $\pm 1^\circ$, $\pm 1.5^\circ$). Dents are sized in millimetres as an inspector would specify them: depth log-uniform in $[0.1, 1.0]$ mm and radius in $[1, 5]$ mm. The case list was hashed before rendering, and no component — shader, threshold or hyperparameter — was refit on this split.

Metrics. Relative local error is the RMS vertex error over the true dent support divided by that of the unchanged CAD (1: no better than CAD; 0: exact). We also report relative whole-airfoil error, depth error, centre error, detection, false physical change on negatives (any vertex moved more than 2×10^{-5}) and abstention on unobservable dents, with 95% bootstrap intervals resampled by blade shape.

5. Experiments

Baselines. Unchanged CAD reports the healthy mesh. Discrete search re-implements our earlier method: image-guided proposals, a fixed cubic profile of radius 0.08 and depths $\{0.0005, 0.001, 0.0015\}$, at nominal pose. Free-form field fits a per-vertex normal displacement with Laplacian smoothness and ℓ_1 sparsity after the same pose and lighting fit. MLP deformation field replaces the per-vertex values with a coordinate MLP (positional encoding, two hidden layers), the continuous deformation field used by CAD-prior reconstruction. These two baselines reimplement the core idea of CADsplat [17] — a smooth deformation field fit to sparse posed views under a CAD prior — within our renderer; they are not the official model, whose code had not been released at the time of writing. 2D detector, lifted is a U-Net trained on (photo, healthy render) pairs of separate exposed blades to segment dents; detections are back-projected onto the mesh through the nominal cameras. All methods share the renderer, cameras and observations, and each is thresholded by the same calibration procedure on its own score.

Main results.

Risk-coverage.

Ablations.

Why recall is limited. Recall is set by dent depth and by the false-alarm budget, not by a failure to locate (Table 5). Every reported dent is in the right place,

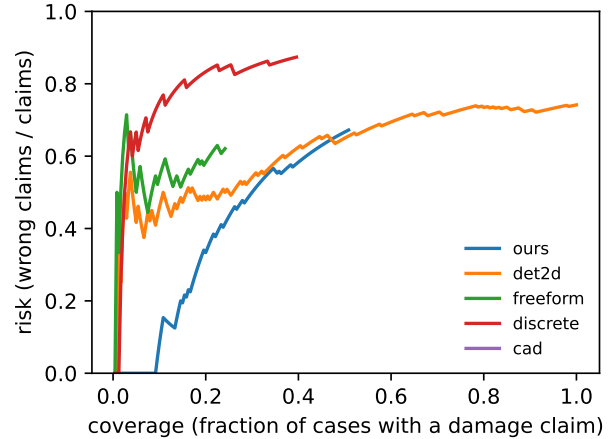


Figure 1. Risk (share of damage claims that are wrong) against coverage (share of cases with a claim) as the threshold varies, sealed test.

and 12/13 visible dents deeper than 0.2mm are reported; most misses are shallower than 0.1 mm, where the change is about one grey level. In 6 of the 17 misses the dent hypothesis converged on the true location, but its evidence stayed below $\tau = 1.52$. The threshold is held up by appearance changes: stains account for 8 of the 9 false alarms, with stain-only fit gains up to 4.7 (5.6 for one healthy blade under camera error). Lowering τ to 0.54 would reach 70% recall at a 20% false-alarm rate. The misses are therefore the price of the false-alarm guarantee: an inspector receives few spurious damage claims, and dents below about 0.1 mm are reported as unsupported rather than guessed.

Transfer to unseen blade designs. Table 6 applies every method, unchanged, to the six CAD6 blades (456 cases). Separation falls from 0.82 on the Rotor37 sealed test to 0.74 (transfer gap 0.08), with 94% of visible dents found, 67% false alarms and 33% abstention on hidden dents. The threshold was calibrated on Rotor37 only, so the false-alarm rate here measures how far the guarantee survives a change of geometry, not a property the method was fit to. Per blade (Table 7), dents are still ranked above negatives (best AUROC on blade 1 (24 mm, AUROC 0.98), worst on blade 3 (31 mm, AUROC 0.64)), but the frozen threshold fails on the smaller blades: false alarms fall with airfoil span (Spearman $\rho = -0.88$, $p = 0.02$). Across blades, airfoil span and detection rate have Spearman $\rho = -0.93$ ($p = 0.01$; six blades, so weak evidence): 99% of visible dents were found on the three smaller blades and 88% on the three larger ones. Because dents are specified in millimetres, the same dent is relatively deeper

Method	Rel. local err. (obs.) ↓	Detection (obs.) % ↑	False change % ↓	Abstain (unobs.) % ↑	Centre err. ↓
BladeDelta (ours)	0.69 [0.59, 0.79]	54 [39, 68]	10 [4, 17]	85 [77, 91]	0.01 [0.00, 0.02]
Free-form field	1.00 [1.00, 1.00]	14 [4, 24]	4 [1, 9]	99 [97, 100]	0.14 [0.04, 0.34]
Discrete search	1.00 [0.99, 1.00]	5 [0, 13]	6 [1, 10]	98 [95, 100]	0.02 [0.02, 0.02]
Unchanged CAD	1.00 [1.00, 1.00]	0 [0, 0]	0 [0, 0]	100 [100, 100]	—

Table 2. Sealed test split. Mean [95% shape-bootstrap CI]. Observable: $\geq 50\%$ of the dent visible in the two input views. Thresholds calibrated on the calibration split (split-conformal, $\alpha = 0.05$).

Views	Observable dents	AUROC ↑	Located % ↑	Method	Rel. local err. ↓	AUROC	Found (%)	Located % ↑
2 views	10/60	0.847	60	Ours, Rotor37 sealed test	0.82 [0.71, 0.90]	54 [39, 68]	54 [39, 68]	54 [39, 68]
4 views	17/60	0.777	41	Ours, Free-form field	0.74 [0.67, 0.84]	94 [89, 98]	70 [65, 75]	70 [65, 75]
7 views	29/60	0.857	62	Discrete search	0.65 [0.58, 0.72]	16 [12, 18]	13 [10, 16]	13 [10, 16]
				Unchanged CAD	0.50 [0.50, 0.50]	0 [0, 0]	0 [0, 0]	0 [0, 0]

Table 3. More views make more dents observable and the evidence clearer.

Variant	AUROC ↑	Located % ↑	Method	Rel. local err. ↓	AUROC	Found (%)	Located % ↑
Full method	0.855	72	Ours v1 (frozen), held-out blades 3–6	0.72 [0.65, 0.90]	90 [85, 95]	67 [62, 72]	67 [62, 72]
No camera (pose) correction	0.679	72	Ours v1.1, held-out blades 3–6	0.95 [0.93, 0.98]	86 [77, 95]	83 [77, 89]	83 [77, 89]
No stain (colour) model	0.753	85	Ours, CAD6 leave-one-blade-out τ	0.70 [0.63, 0.84]	13 [3, 26]	6 [1, 12]	6 [1, 12]
Whole image instead of interior evidence	0.685	61					
No score-test proposals	0.420	72					
No low-frequency brightness field	0.846	72					

Table 4. Ablations on the development subset (18 observable dents, 18 healthy or stained blades).

True depth (mm)	2 views	2 views (same cases)	4 views	7 views	Cases	Visible dents	AUROC	Found (%)
< 0.1	1/10 (1 loc.)	0/4 (0 loc.)	1/5 (0 loc.)	3/26 (3 loc.)	76	31	0.98	100
0.1–0.2	7/14 (7 loc.)	1/2 (1 loc.)	2/6 (2 loc.)	7/11 (7 loc.)	76	31	0.72	100
> 0.2	12/13 (12 loc.)	4/4 (4 loc.)	4/5 (4 loc.)	8/9 (6 loc.)	76	31	0.64	97
All visible dents	20/37 (20 loc.)	5/10 (5 loc.)	7/17 (6 loc.)	18/20 (16 loc.)	76	31	0.66	94
					76	31	0.93	84
					76	31	0.86	87

Table 5. Visible test dents reported (located within 0.1 span) by true depth, at the frozen threshold. Below 0.1 mm a dent changes the image by about one grey level; the method declines rather than guess.

Table 6. Transfer to six unseen real blade designs (CAD6, 456 cases; dents 0.1–1.0 mm deep). Every method is frozen: the shader trained on Rotor37 only, thresholds from the Rotor37 calibration split. Top row: ours on the Rotor37 sealed test, for the transfer gap. Brackets: 95% intervals resampling blades.

Blade	Airfoil span (mm)	Cases	Visible dents	AUROC	Found (%)
1	24	76	31	0.98	100
2	26	76	31	0.72	100
3	31	76	31	0.64	97
4	40	76	31	0.66	94
5	52	76	31	0.93	84
6	70	76	31	0.86	87

Table 7. Ours on each CAD6 blade, smallest to largest airfoil (frozen threshold).

on a small blade; detection tracks depth relative to the airfoil span ($\rho = 0.10$) more strongly than depth in millimetres ($\rho = 0.02$), rising from 92% in the shallowest relative-depth tertile to 97% in the deepest. Diagnosis and fix (v1.1). On a healthy CAD6 capture whose image is identical to the reference, the silhouette pose fit moves a correct camera by about 0.3° and a spurious dent absorbs the misregistration; under real camera error, the image-space reference bias — large on unseen geometry because the shader was trained on Rotor37 only — stays fixed while the blade moves. v1.1 corrects both: it applies the reference correction to the silhou-

ette as well as to shading, and it reprojects the shading correction through the surface so that it moves with the blade (identical to v1 at the nominal pose). Both changes were developed on CAD6 blades 1–2 only and frozen, with a threshold recalibrated on the Rotor37 calibration split, before blades 3–6 were run. On those held-out blades v1.1 raises AUROC from 0.72 to 0.95, cuts false alarms from 51% to 4%, locates 83% of visible dents (v1: 67%) and abstains on 98% of hidden dents (v1: 50%). Recalibrating the threshold on CAD6 negatives, leave-one-blade-out, gives 13% found at 4% false alarms: the cost of transferring a threshold across blade designs.

Variant (CAD6, 48 cases)	AUROC	Found (%)	False alarms / defects / negatives	6 / 22 / 31
Full method	0.71 [0.62, 0.83]	92 [83, 100]	78 [56, 100]	0.540
No camera correction	0.90 [0.81, 0.98]	92 [83, 100]	56 [33, 67]	0/22
No stain model	0.73 [0.64, 0.84]	96 [88, 100]	89 [78, 100]	0/22
			False alarms	1/31

Table 8. Ablations on CAD6 (frozen settings, fixed 48-case subset).

Table 9. Real objects (SiM3D) scanned before and after damage; truth from the measured scans.

A pre-registered negative result (v2). Targeting the remaining false alarms, we pre-registered three changes and a rule on development blades: a smooth surface appearance field shared by both hypotheses (A), a dent-size prior (D), or both (E), adopted only if recall at matched false-alarm rate improved by at least five points. None did: D matched v1 (67% recall, AUROC 0.87 vs. 0.85), while A and E raised negative fit gains so much that recall fell to 6%; the extra appearance freedom helped the dent hypothesis as much as the null. The change that does reduce false alarms is v1.1’s reference correction of silhouette and shading (above), which also lowers the worst development false-alarm gains on Rotor37 (2.46 to 0.09).

Physical validation (pending real validation).

Pending real validation. A $2\times$ print of CAD6 blade 5 will be dented with steel balls (target depths 0.3, 0.7 and 1.2 mm), photographed with marker-based poses before and after denting, and measured by structured-light scan (or a 0.01 mm dial gauge). This section is reserved for those results; the rest of the paper does not depend on it.

Qualitative results and failures. [TODO: Figure: observed images, H_0 residual, recovered vs. true dent for successes, a colour/dent ambiguity case and an unobservable dent.]

Real objects. No public dataset pairs real blade photographs with measured damage geometry, so we use SiM3D [7]: six wooden stools scanned with an industrial structured-light sensor before and after damage, with calibrated photographs and expert defect annotations. The healthy scan replaces the CAD model, the healthy photograph from the same scanner position serves as the reference after 2D alignment, and the measured scan difference is the truth; annotated regions with no measured shape change (paint, contamination) count as negatives. On these textured objects our method does not separate physical defects from undamaged areas (Table 9). The residual between sessions is dominated by wood-grain misregistration and

specular changes that the shader, trained on 12 photographs per object, does not model. We report this as a failure case: the evidence test is only as reliable as the appearance model behind it.

6. Limitations and Conclusion

Our positive results use rendered images of public compressor geometry with synthetic damage. On real scanned objects (SiM3D) the method did not separate physical defects from undamaged areas; real-blade validation with measured damage remains future work, and an appearance model that holds on real materials is its main prerequisite. The dent model has one dent per blade, and edge damage such as nicks and material loss is out of scope. The conformal guarantee holds for negatives exchangeable with the calibration set; a new camera rig or lighting setup requires recalibration. The method answers whether these views support a physical change; it does not certify airworthiness or replace an inspector.

[TODO: conclusion paragraph after results]

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